NYC Taxi Operations Intelligence Platform

Phase 0 Summary and Interview Preparation

Phase 0 Summary

Phase 0 prepared the project foundation before any ingestion, cleaning, dashboarding, dbt, testing, or CI/CD work. The goal was to make sure the local repo, Python environment, Google Cloud setup, BigQuery workspace, and GitHub backup were ready before moving into data engineering work.

Completed Item	Plain-English Meaning
Project repository created	The project now has a local home on your Mac.
Starter folder structure created	Files are organized before code and data start growing.
Documentation files created	The project story, roadmap, architecture, and setup notes are written down.
Python virtual environment created	Python packages for this project stay separate from the rest of the Mac.
Python packages installed	The project has tools for data files, BigQuery, environment variables, and progress bars.
Google Cloud connected	Your Mac can communicate with your GCP account.
BigQuery dataset created	The cloud warehouse has an empty container ready for future tables.
First Git commit created	The setup was saved as a local history checkpoint.
GitHub push completed	The project is backed up online and visible in GitHub.

Current Project Details

Item	Value
GitHub repository	https://github.com/ilyassfcb11-lgtm/-nyc-taxi-intelligence.git
GCP project ID	nyc-taxi-project-502819
BigQuery dataset	nyc-taxi-project-502819.nyc_taxi_ops
Latest Git commit	ee073f6 Create Phase 0 project setup
Current phase status	Phase 0 complete. Ready for Phase 1 ingestion.

What Each Tool Is For

Tool	Purpose
Git	Tracks project history on the Mac.
GitHub	Stores the repository online so recruiters and collaborators can view it.
Python	Will run ingestion scripts, data utilities, and validation checks.
Virtual environment	Keeps this project's Python packages isolated from other projects.
Google Cloud Platform	Provides the cloud environment for BigQuery.
BigQuery	Cloud data warehouse for raw data, cleaned tables, and KPI marts.
Tableau	Dashboard and visualization tool for the business-facing analysis.

Files Created

File	Purpose
README.md	The project front page. It explains the project and current phase.
PROJECT_BRIEF.md	The business framing and objective.
ROADMAP.md	The phased plan from setup to ingestion, dashboards, dbt, testing, and polish.
ARCHITECTURE.md	The high-level system flow from NYC TLC files to Tableau dashboards.
COMMANDS.md	A command cheat sheet so terminal usage is documented and reusable.
docs/GCP_SETUP.md	The Google Cloud and BigQuery setup notes.
requirements.txt	The Python package list.
.gitignore	Rules for keeping local data, secrets, and temporary files out of GitHub.
data/README.md	Explains how local data files will be handled later.
scripts/README.md	Placeholder for future Python scripts.
sql/README.md	Placeholder for SQL cleaning, modeling, and KPI queries.
docs/README.md	Placeholder for supporting documentation.

Scope And Timing Check

Phase 0 is complete and the project is still on track. The target deliverable was repo, environment, and GCP readiness. That deliverable is complete. Phase 1 should begin with a small documentation step: create DATA_SOURCES.md before downloading any data.

Interview Questions And Practice Answers

1. What was the goal of Phase 0?

The goal was to set up the project foundation before writing ingestion or analytics code. I created the repository structure, Python environment, documentation, Google Cloud connection, BigQuery dataset, and first GitHub checkpoint.

2. Why did you use Git?

I used Git to track the history of the project. Each commit acts like a checkpoint, so I can see what changed over time and keep the project organized professionally.

3. What is the difference between Git and GitHub?

Git is the local version-control tool on my Mac. GitHub is the online platform where I publish the repository so others, including recruiters, can view it.

4. Why did you create a Python virtual environment?

A virtual environment keeps the project's Python packages isolated. That means this project can use its own package versions without interfering with other Python projects on my computer.

5. Why did you create a BigQuery dataset before loading data?

The dataset is the container where project tables will live. Creating it first gives the project a clean warehouse structure before ingestion begins.

6. What is BigQuery's role in this project?

BigQuery is the cloud data warehouse. Raw taxi data will be loaded there, then SQL transformations will create cleaned tables and KPI marts for Tableau.

Interview Questions And Practice Answers

7. Why is documentation important in this project?

Documentation makes the project easier to understand, reproduce, and discuss in interviews. It shows the business context, architecture, roadmap, setup decisions, and later KPI logic.

8. How did you control project scope in Phase 0?

I focused only on setup. I did not start ingestion, cleaning, dashboarding, dbt, or CI/CD yet. This keeps the project organized and easier to learn step by step.

9. How does this phase support Data Engineering?

It prepares the environment for data engineering work: Python scripts, cloud access, BigQuery setup, and repeatable project structure.

10. How does this phase support Analytics Engineering and Data Analysis?

It creates the foundation for later SQL models, KPI marts, Tableau dashboards, and business analysis. The setup makes sure future work is organized and reproducible.

Next Phase

Phase 1 is ingestion. The first step should be documenting the exact official NYC TLC Yellow Taxi files in DATA_SOURCES.md. After that, we will write a download script and load raw data into BigQuery.